UIT2521 – Information Theory and Applications

UNIT II INFORMATION THEORY FUNDAMENTALS

Tutorial – IV

Extension of Discrete Memoryless Source

1. A zero memory source contains X = {x_1, x_2, x_3, x_4} with P(X) = {1/2, 1/4, 1/8, 1/8}
   a) Determine the entropy of the source.
   b) Determine the second order extension of the source. Show that $H(X^2) = 2H(X)$.

2. For DMS 'X' with two symbols x_1 and x_2 and $p(x_1) = 0.9$ and $p(x_2) = 0.1$. Find out the second order extension for the source. Find the efficiency η and redundancy of the code.

1. Zero Memory Source: X = {x_1, x_2, x_3, x_4}, P(X) = {1/2, 1/4, 1/8, 1/8}

(a) Entropy of the Source

The source entropy is:

$$H(X) = -\sum_{i=1}^{4} p(x_i) \log_2 p(x_i)$$

Calculate each symbol's contribution:

- $p(x_1) = 1/2$: $-\frac{1}{2}\log_2(\frac{1}{2}) = 0.5 \times 1 = 0.5$
- $p(x_2) = 1/4$: $-\frac{1}{4}\log_2(\frac{1}{4}) = 0.25 \times 2 = 0.5$
- $p(x_3) = 1/8$: $-\frac{1}{8}\log_2(\frac{1}{8}) = 0.125 \times 3 = 0.375$
- $p(x_4) = 1/8$: $-\frac{1}{8}\log_2(\frac{1}{8}) = 0.125 \times 3 = 0.375$

Add them:

$$H(X) = 0.5 + 0.5 + 0.375 + 0.375 = \boxed{1.75 \text{ bits}}$$

(b) Second Order Extension and $H(X^2) = 2H(X)$

Second order extension: All pairs of symbols, i.e., the source emits two symbols at a time. There are $4 \times 4 = 16$ symbol pairs. For a zero-memory source, the probability of any pair (x_i, x_j) is $p(x_i)p(x_j)$.

The entropy for the second order extension:

$$H(X^2) = -\sum_{i=1}^{4}\sum_{j=1}^{4} p(x_i)p(x_j) \log_2[p(x_i)p(x_j)]$$

Expand the logarithm:

$$\log_2[p(x_i)p(x_j)] = \log_2 p(x_i) + \log_2 p(x_j)$$

So,

$$H(X^2) = -\sum_{i=1}^{4}\sum_{j=1}^{4} p(x_i)p(x_j)\left[\log_2 p(x_i) + \log_2 p(x_j)\right]$$

This splits as:

This splits as:

$$= -\sum_i p(x_i) \log_2 p(x_i) \sum_j p(x_j) - \sum_j p(x_j) \log_2 p(x_j) \sum_i p(x_i)$$

Since $\sum_k p(x_k) = 1$, both terms are $H(X)$:

$$H(X^2) = 2H(X)$$

So for this example:

$$H(X^2) = 2 \times 1.75 = \boxed{3.5 \text{ bits}}$$

This always holds for a zero-memory source ⊡ 1 B .

2. DMS X: Two symbols x_1 and x_2, $p(x_1) = 0.9$, $p(x_2) = 0.1$

(a) Second Order Extension

Source alphabet: $\{x_1, x_2\}$, Probabilities: $p(x_1) = 0.9$, $p(x_2) = 0.1$

Second order extension: All 4 possible pairs, with probabilities:

- (x_1, x_1): $0.9 \times 0.9 = 0.81$
- (x_1, x_2): $0.9 \times 0.1 = 0.09$
- (x_2, x_1): $0.1 \times 0.9 = 0.09$
- (x_2, x_2): $0.1 \times 0.1 = 0.01$

So, for the new source X^2, the symbol set is $\{(x_1, x_1), (x_1, x_2), (x_2, x_1), (x_2, x_2)\}$, and the probabilities are $0.81, 0.09, 0.09, 0.01$.

Entropy:

$$\begin{aligned} H(X) &= -[0.9 \log_2 0.9 + 0.1 \log_2 0.1] \\ &= -[0.9 \times (-0.152) + 0.1 \times (-3.322)] \\ &= 0.1368 + 0.3322 = \boxed{0.469 \text{ bits}} \end{aligned}$$

Second order entropy:

Second order entropy:

$$H(X^2) = 2H(X) = 2 \times 0.469 = \boxed{0.938 \text{ bits}}$$

(b) Efficiency (η) and Redundancy

Maximum entropy for binary source: $H_\text{max} = 1\text{ bit}$ (when $p=0.5$).

Efficiency:

$$\eta = \frac{H(X)}{H_{\max}} = \frac{0.469}{1} = 46.9\%$$

Redundancy:

$$\text{Redundancy} = 1 - \eta = 1 - 0.469 = 0.531 = 53.1\%$$

Hint:

Extension of Discrete Memoryless Source:

The entropy of the extension of discrete memoryless source

$$\boxed{\mathbf{H(S^n) = n\,H(S)} \text{ bits/ symbol}}$$

Where $\mathbf{S^n}$ is the **extended source alphabet** with $\mathbf{K^n}$ **distinct symbols** and **n** is the **order of extension**.

******